

Now, we consider the permutation-based procedure. For a square matrix A , we let A^0 denote the matrix A with its diagonal entries set equal to 0. We have

$$\begin{aligned} \widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n) &= \frac{1}{m(m-1)} \sum_{1 \leq i \neq i' \leq m} k(X_i, X_{i'}) + \frac{1}{n(n-1)} \sum_{1 \leq j \neq j' \leq n} k(Y_j, Y_{j'}) \\ &\quad - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(X_i, Y_j) \\ &= \left(K_\lambda^0 \circ \begin{pmatrix} \mathbb{1}_m \mathbb{1}_m^\top / (m(m-1)) & -\mathbb{1}_m \mathbb{1}_n^\top / (mn) \\ -\mathbb{1}_n \mathbb{1}_m^\top / (mn) & \mathbb{1}_n \mathbb{1}_n^\top / (n(n-1)) \end{pmatrix} \right)_+ \end{aligned}$$

where it is not possible to rewrite the matrix as an outer product of vectors. Instead, we break it down into a sum of three outer products of vectors.

$$\begin{aligned} \begin{pmatrix} \mathbb{1}_m \mathbb{1}_m^\top / (m(m-1)) & -\mathbb{1}_m \mathbb{1}_n^\top / (mn) \\ -\mathbb{1}_n \mathbb{1}_m^\top / (mn) & \mathbb{1}_n \mathbb{1}_n^\top / (n(n-1)) \end{pmatrix} &= \left(\frac{1}{m(m-1)} - \frac{1}{mn} \right) \begin{pmatrix} \mathbb{1}_m \mathbb{1}_m^\top & \mathbb{0}_m \mathbb{0}_n^\top \\ \mathbb{0}_n \mathbb{0}_m^\top & \mathbb{0}_n \mathbb{0}_n^\top \end{pmatrix} \\ &\quad + \left(\frac{1}{n(n-1)} - \frac{1}{mn} \right) \begin{pmatrix} \mathbb{0}_m \mathbb{0}_m^\top & \mathbb{0}_m \mathbb{0}_n^\top \\ \mathbb{0}_n \mathbb{0}_m^\top & \mathbb{1}_n \mathbb{1}_n^\top \end{pmatrix} \\ &\quad + \frac{1}{mn} \begin{pmatrix} \mathbb{1}_m \mathbb{1}_m^\top & -\mathbb{1}_m \mathbb{1}_n^\top \\ -\mathbb{1}_n \mathbb{1}_m^\top & \mathbb{1}_n \mathbb{1}_n^\top \end{pmatrix} \\ &= \frac{n-m+1}{mn(m-1)} uu^\top + \frac{m-n+1}{mn(n-1)} ww^\top + \frac{1}{mn} vv^\top \end{aligned}$$

where $u := (\mathbb{1}_m, \mathbb{0}_n)$, $w := (\mathbb{0}_m, -\mathbb{1}_n)$ and $v := (\mathbb{1}_m, -\mathbb{1}_n)$ of shapes $(m+n) \times 1$, with $\mathbb{0}_n$ denoting the vector of length n with all entries equal to 0. Using this fact, we obtain that

$$\widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m, \mathbb{Y}_n) = \frac{n-m+1}{mn(m-1)} u^\top K_\lambda^0 u + \frac{m-n+1}{mn(n-1)} w^\top K_\lambda^0 w + \frac{1}{mn} v^\top K_\lambda^0 v.$$

For a vector $a = (a_1, \dots, a_\ell)$ and a permutation $\tau: \{1, \dots, \ell\} \rightarrow \{1, \dots, \ell\}$, we denote the permuted vector as $a_\tau = (a_{\tau(1)}, \dots, a_{\tau(\ell)})$. Recall that $U_i := X_i$, $i = 1, \dots, m$ and $U_{m+j} := Y_j$, $j = 1, \dots, n$. Consider a permutation $\sigma: \{1, \dots, m+n\} \rightarrow \{1, \dots, m+n\}$ and let $\mathbb{X}_m^\sigma := (U_{\sigma(i)})_{1 \leq i \leq m}$ and $\mathbb{Y}_n^\sigma := (U_{\sigma(m+j)})_{1 \leq j \leq n}$. Following a similar reasoning to the one presented in Appendix B, we find

$$\begin{aligned} \widehat{M}_\lambda^\sigma &:= \widehat{\text{MMD}}_{\lambda, \mathbf{a}}^2(\mathbb{X}_m^\sigma, \mathbb{Y}_n^\sigma) \\ &= \frac{n-m+1}{mn(m-1)} u_{\sigma^{-1}}^\top K_\lambda^0 u_{\sigma^{-1}} + \frac{m-n+1}{mn(n-1)} w_{\sigma^{-1}}^\top K_\lambda^0 w_{\sigma^{-1}} + \frac{1}{mn} v_{\sigma^{-1}}^\top K_\lambda^0 v_{\sigma^{-1}} \end{aligned}$$

since $\{k_\lambda(X_i, Y_i) : i = 1, \dots, n\} = \{k_\lambda(X_{\sigma(i)}, Y_{\sigma(i)}) : i = 1, \dots, n\}$. The aim is to compute $(\widehat{M}_\lambda^{\sigma^{(b)}})_{1 \leq b \leq B}$ efficiently for any $B \in \mathbb{N} \setminus \{0\}$. We let U , V and W denote the $(m+n) \times B$ matrices of stacked vectors $(u_{\sigma^{(b)-1}})_{1 \leq b \leq B}$, $(v_{\sigma^{(b)-1}})_{1 \leq b \leq B}$ and $(w_{\sigma^{(b)-1}})_{1 \leq b \leq B}$, respectively. We are then able to compute $(\widehat{M}_\lambda^{\sigma^{(b)}})_{1 \leq b \leq B}$ as

$$\begin{aligned} &\frac{n-m+1}{mn(m-1)} \text{diag}(U^\top K_\lambda^0 U) + \frac{m-n+1}{mn(n-1)} \text{diag}(W^\top K_\lambda^0 W) + \frac{1}{mn} \text{diag}(V^\top K_\lambda^0 V) \\ &= \frac{n-m+1}{mn(m-1)} \sum_{\text{rows}} U \circ (K_\lambda^0 U) + \frac{m-n+1}{mn(n-1)} \sum_{\text{rows}} W \circ (K_\lambda^0 W) + \frac{1}{mn} \sum_{\text{rows}} V \circ (K_\lambda^0 V). \end{aligned}$$

Since the inverse map for permutations is a bijection between the space of all permutations and itself, it follows that uniformly generating B permutations and taking their inverses is equivalent to directly uniformly generating B permutations. So, in practice, we can simply uniformly generate permutations $\tau^{(1)}, \dots, \tau^{(B)}$ and assume that these correspond to $\sigma^{(1)-1}, \dots, \sigma^{(B)-1}$ for uniformly generated permutations $\sigma^{(1)}, \dots, \sigma^{(B)}$. We can now present an efficient version for Step 1 of Algorithm 1 using permutations.

Efficient Step 1 using permutations:

construct $(m+n) \times (B_1 + B_2 + 1)$ matrix U of stacked vectors of $(\mathbb{1}_m, \mathbb{0}_n)$
 construct $(m+n) \times (B_1 + B_2 + 1)$ matrix V of stacked vectors of $(\mathbb{1}_m, -\mathbb{1}_n)$
 construct $(m+n) \times (B_1 + B_2 + 1)$ matrix W of stacked vectors of $(\mathbb{0}_m, -\mathbb{1}_n)$
 use $B_1 + B_2$ permutations to permute the elements of the columns of U , V and W without permuting the elements of the $(B_1 + 1)^{\text{th}}$ columns of U , V and W
for $\lambda \in \Lambda$:
 compute kernel matrix K_λ^0 with zero diagonals
 compute $(\widehat{M}_{\lambda,1}^1, \dots, \widehat{M}_{\lambda,1}^{B_1+1}, \widehat{M}_{\lambda,2}^1, \dots, \widehat{M}_{\lambda,2}^{B_2})$ as

$$\frac{n-m+1}{mn(m-1)} \sum_{\text{rows}} U \circ (K_\lambda^0 U) + \frac{m-n+1}{mn(n-1)} \sum_{\text{rows}} W \circ (K_\lambda^0 W) + \frac{1}{mn} \sum_{\text{rows}} V \circ (K_\lambda^0 V)$$

$$(\widehat{M}_{\lambda,1}^{\bullet 1}, \dots, \widehat{M}_{\lambda,1}^{\bullet B_1+1}) = \text{sort_by_ascending_order}(\widehat{M}_{\lambda,1}^1, \dots, \widehat{M}_{\lambda,1}^{B_1+1})$$

Appendix D. Lower bound on the minimax rate over a Sobolev ball

We claim that the two-sample minimax rate of testing over the Sobolev ball $\mathcal{S}_d^s(R)$ is $n^{-2s/(4s+d)}$. Formally, let $\alpha, \beta \in (0, 1)$, $d \in \mathbb{N} \setminus \{0\}$ and $M, s, R \in (0, \infty)$, we claim that there exists some positive constant $C'_0(M, d, s, R, \alpha, \beta)$ such that

$$\underline{\rho}(\mathcal{S}_d^s(R), \alpha, \beta, M) := \inf_{\Delta_\alpha} \rho(\Delta_\alpha, \mathcal{S}_d^s(R), \beta, M) \geq C'_0(M, d, s, R, \alpha, \beta) n^{-2s/(4s+d)} \quad (18)$$

where the infimum is taken over all tests Δ_α of non-asymptotic level α and where $c' \leq \frac{m}{n} \leq C'$ for some positive constants c' and C' .

The proof mirrors the reasoning of Albert et al. (2022, Section 4, Theorem 4) who derive the independence minimax rate of testing $n^{-2s/(4s+d_1+d_2)}$ over the Sobolev ball $\mathcal{S}_{d_1+d_2}^s(R)$ considering paired samples on $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$. While the case $d = 1$ is not covered by their framework, the proof extends naturally to it. We illustrate the main reasoning behind the proof as this is of interest for our experiments in Section 5.5.

First, note that the minimax rate $\underline{\rho}(\mathcal{S}_d^s(R), \alpha, \beta, M)$ is

$$\begin{aligned} \inf_{\Delta_\alpha} \rho(\Delta_\alpha, \mathcal{S}_d^s(R), \beta, M) &= \inf_{\Delta_\alpha} \inf \left\{ \tilde{\rho} > 0 : \sup_{(p,q) \in \mathcal{F}_{\tilde{\rho}}^M(\mathcal{S}_d^s(R))} \mathbb{P}_{p \times q}(\Delta_\alpha(\mathbb{X}_m, \mathbb{Y}_n) = 0) \leq \beta \right\} \\ &= \inf \left\{ \tilde{\rho} > 0 : \inf_{\Delta_\alpha} \sup_{(p,q) \in \mathcal{F}_{\tilde{\rho}}^M(\mathcal{S}_d^s(R))} \mathbb{P}_{p \times q}(\Delta_\alpha(\mathbb{X}_m, \mathbb{Y}_n) = 0) \leq \beta \right\} \end{aligned}$$

where $\mathcal{F}_{\tilde{\rho}}^M(\mathcal{S}_d^s(R)) := \{(p, q) : \max(\|p\|_\infty, \|q\|_\infty) \leq M, p - q \in \mathcal{S}_d^s(R), \|p - q\|_2 > \tilde{\rho}\}$. Hence, to prove Equation (18) it suffices to construct two probability densities p and q on $\mathbb{R}^d$ which satisfy $\max(\|p\|_\infty, \|q\|_\infty) \leq M$, $p - q \in \mathcal{S}_d^s(R)$, $\|p - q\|_2 < C'_1(m + n)^{-2s/(4s+d)}$ for some $C'_1 > 0$, and for which $\mathbb{P}_{p \times q}(\Delta'_\alpha(\mathbb{X}_m, \mathbb{Y}_n) = 0) > \beta$ holds for all tests Δ_α with non-asymptotic level α . Intuitively, one constructs densities which are close enough in L^2 -norm so that any test with non-asymptotic level α fails to distinguish them from one another.

Albert et al. (2022, Section 4) show that a suitable choice of p and q is to take the uniform probability density on $[0, 1]^d$ and a perturbed version of it. To construct the latter, first define for all $u \in \mathbb{R}$ the function

$$G(u) := \exp\left(-\frac{1}{1 - (4u + 3)^2}\right) \mathbb{1}_{(-1, -\frac{1}{2})}(u) - \exp\left(-\frac{1}{1 - (4u + 1)^2}\right) \mathbb{1}_{(-\frac{1}{2}, 0)}(u)$$

which is plotted in Figure 2 in Section 5.5. Consider $P \in \mathbb{N} \setminus \{0\}$ and some vector $\theta = (\theta_\nu)_{\nu \in \{1, \dots, P\}^d} \in \{-1, 1\}^{P^d}$ of length P^d with entries either -1 or 1 which is indexed by the P^d d -dimensional elements of $\{1, \dots, P\}^d$. Then, the perturbed uniform density is defined as

$$f_\theta(u) := \mathbb{1}_{[0, 1]^d}(u) + P^{-s} \sum_{\nu \in \{1, \dots, P\}^d} \theta_\nu \prod_{i=1}^d G(Pu_i - \nu_i) \quad (19)$$

for all $u \in \mathbb{R}^d$. As illustrated in Figure 2, this indeed corresponds to a uniform probability density with P perturbations along each dimension.

This construction by Albert et al. (2022) is a generalisation of the detailed construction of Butucea (2007, Section 5) for the 1-dimensional case for goodness-of-fit testing. We also point out the work of Li and Yuan (2019, Theorems 5, part (ii)) who use a similar construction to show that, for any alternative¹⁰ in $\mathcal{F}_{\tilde{\rho}}^M(\mathcal{S}_d^s(R))$ with rate $\tilde{\rho}$ smaller or equal to $n^{-2s/(4s+d)}$, there exists some $\alpha \in (0, 1)$ such that any test with asymptotic level α must have power asymptotically strictly smaller than one. The original idea behind all those constructions is due to Ingster (1987, 1993b) with his work on nonparametric minimax rates for goodness-of-fit testing.

Appendix E. Proofs

In this section, we prove the statements presented in Section 3. We first introduce some standard results.

First, recall that we assume $m \leq n$ and $n \leq Cm$ for some constant $C \geq 1$ as in Equation (7). It follows that

$$\frac{1}{m} + \frac{1}{n} \leq \frac{C + 1}{n} = \frac{2(C + 1)}{2n} \leq \frac{2(C + 1)}{m + n} \quad \text{and} \quad \frac{1}{m + n} \leq \frac{2}{m + n} \leq \frac{1}{m} + \frac{1}{n}. \quad (20)$$

10. To be more precise, the alternatives considered by Li and Yuan (2019) require both p and q to belong to the Sobolev ball (rather than just $p - q$ in our case) but do not require these densities to be bounded. Moreover, they use the non-homogeneous Sobolev space definition (with an extra ‘+1’ in the weighting term) while we use the homogeneous definition (without that extra term).